Hyperspectral Reconstruction from Spatial-Spectral Coded Images

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Motivation and Related Work

Hyperspectral imaging has demonstrated its utility in a multitude of applications, such as material classification, remote sensing, biomedical diagnosis, and image segmentation. Conventional approaches to hyperspectral imaging make use of narrowband light sources or tunable optical-bandpass filters to sequentially scan the scene and build a spectral cube. Other approaches like push broom or line-scan cameras capture a spectral cube with one focal plane array (FPA) measurement per spatial line of the scene. The major disadvantage of these techniques is the poor temporal resolution of the image acquisition process, which deteriorates linearly in proportion to the desired spatial and spectral resolution. This is especially important for handheld devices. Snapshot hyperspectral cameras do exist; however, their practical use is limited by their high costs ($20K+).

In order to make hyperspectral imaging systems practical, we need to make them accessible (in terms of their availability and cost), and usable (by optimizing the data acquisition process). Coded aperture snapshot imagers (CASSI) [1] which embody the principles of compressive sensing, do a remarkable job at constructing the spectral cube with a few or even one FPA measurement. This solves the problem of optimizing for the data acquisition process; however, the requirement of a coded aperture mask and collimating optics limits the ubiquity of this approach. Alternatively, the rolling shutter sampling, which is abundantly available in commercial and machine vision cameras, provides an inherent spatial and temporal coding that could be leveraged for compressive sensing applications [2]. Prior work [3] has demonstrated the use of rolling shutter cameras for compressive temporal imaging, i.e., capturing video from a still image. In this project, we plan to leverage similar principles, however, for compressive spectral imaging.

Project Overview

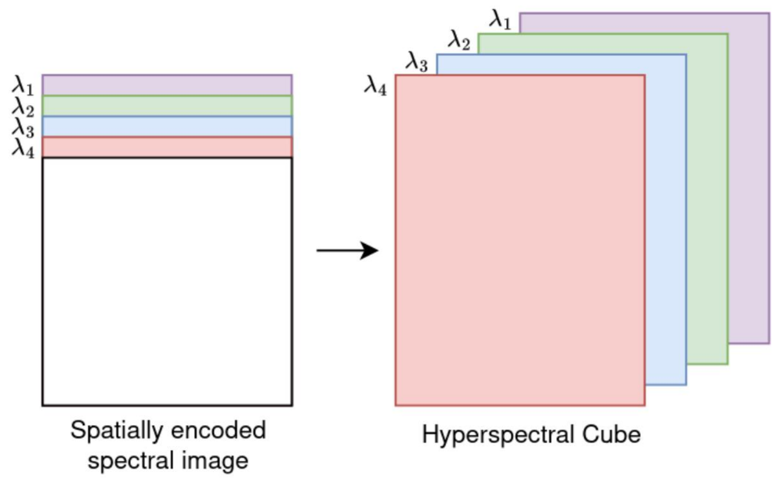
Capturing hyperspectral images using rolling shutter cameras would require two primary steps, viz; leveraging the rolling shutter mechanism to capture a spatial-spectral coded image (*acquisition step*) and transforming the spatial-spectral image into the spectral cube (*reconstruction step*).

Acquisition Step

In this step, one could sequentially illuminate the scene with light sources of different wavelengths, either by using multiple narrow-band light sources or a tunable optical filter. The exposure time, however, has to be adjusted such that for different read operations, as the rolling shutter progresses, spectrally distinct pixel rows are captured (see figure for illustration). In the interest of time, this step will not be implemented on hardware for this project, and instead, we will use hyperspectral datasets [CAVE, ICVL, ARAD-HS] to simulate the image acquisition. We will test multiple encoding schemes to simulate the spatial-spectral images and analyze the effect on different reconstruction algorithms.

Reconstruction Step

Once we have access to spatial-spectral encoded images, we need to interpolate the spatial information for each spectral channel. In this project, we plan to develop algorithms for reconstructing the spectral cube from the spatial-spectral image. Prior work [4] has shown that convolutional neural networks (CNNs) could be used for high-quality reconstruction of hyperspectral images from encoded images. Therefore, we will primarily focus on CNN-based methods and compare them against ADMM-based reconstruction.



Milestones, Timeline & Goals

Goals:

1. Develop hyperspectral reconstruction algorithms (CNN-based)
2. Compare it with baseline/conventional ADMM-based reconstruction
3. Optimize the algorithm by empirically evaluating the different spatial-spectral encoding schemes that follow rolling shutter sampling:
    a. By varying the number of consequent rows per spectral channel
    b. By varying the order of spectral channels (in-order vs. random)

Timeline:

Week 10	Reading the paper [4] and creating a simulated image dataset.
Week 11	Implementing ADMM-based method (baseline) and preparing multiple spatial-spectral encoding schemes
Week 12	Implementing CNN-based reconstruction algorithm and evaluate performance.
Week 13	Algorithm optimization, final experiments, and report/poster writing.

Reference

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